

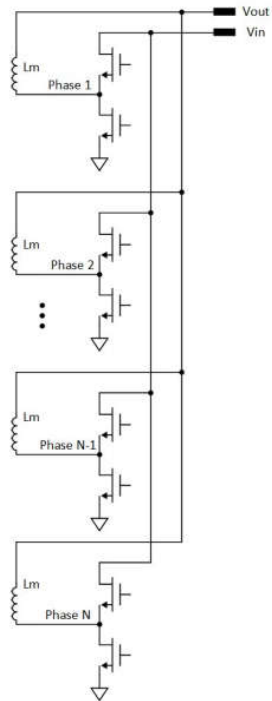
RENESAS INFRASTRUCTURE POWER

TLVR TOPOLOGY

INFRASTRUCTURE POWER
JUNE 2020

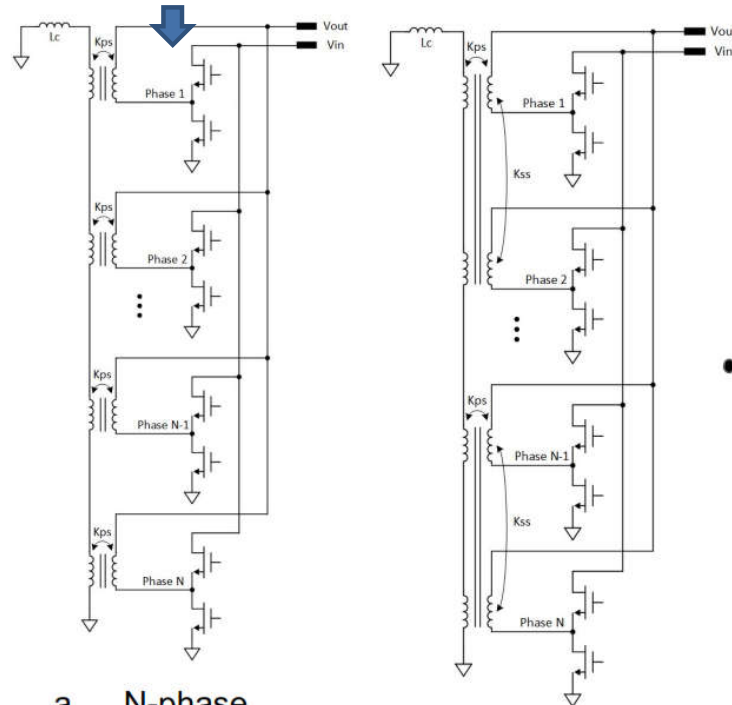
BIG IDEAS
FOR EVERY SPACE

TLVR (TRANS INDUCTOR VR) CONCEPT OVERVIEW



Standard Multiphase
Buck topology

Renesas Focus



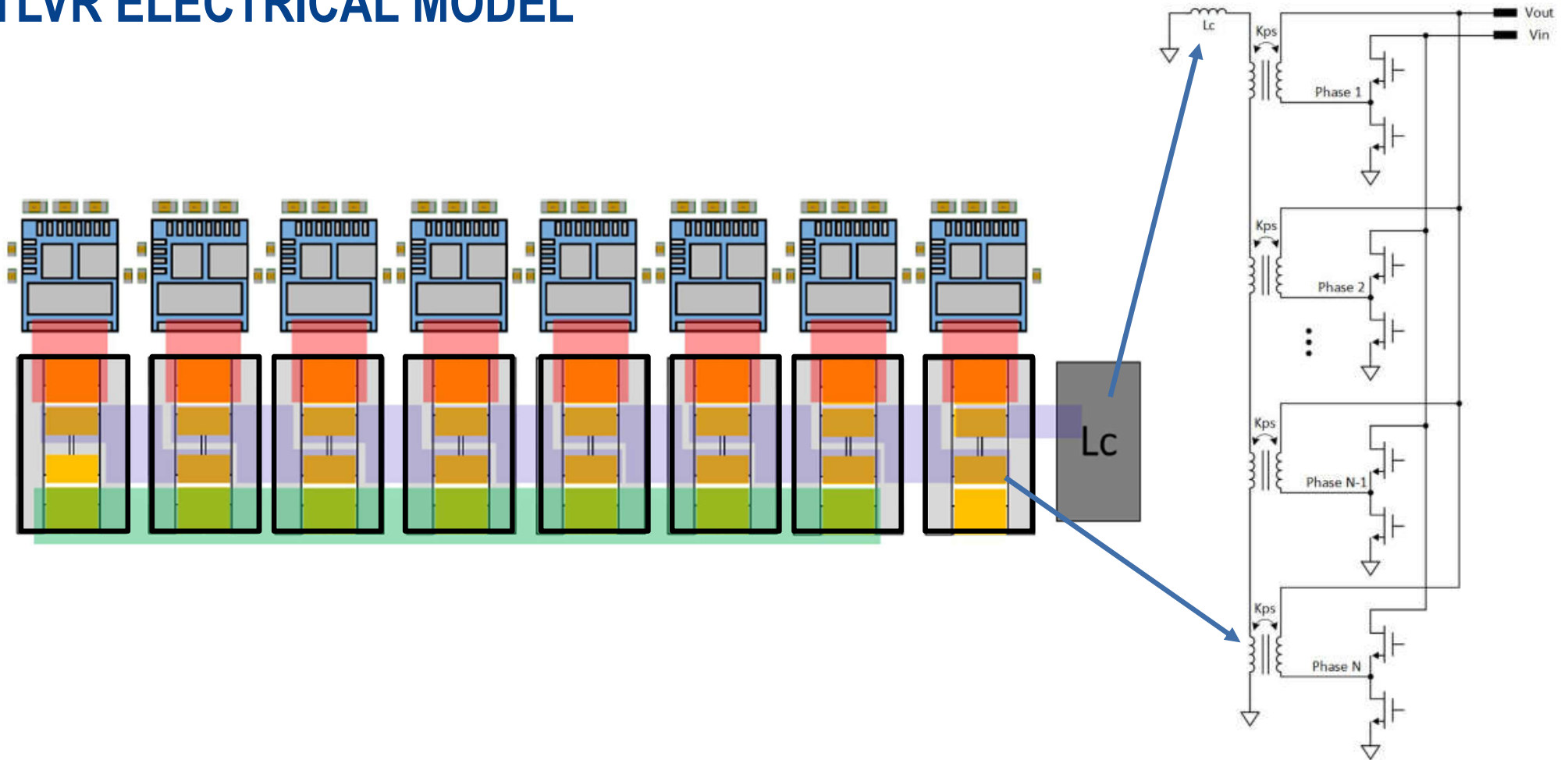
a. N-phase
single-secondary TLVR

b. N-phase
dual-secondary TLVR

- ❑ Each phase has an output inductor which carries DC and magnetizing ripple current
- ❑ Each output inductor winding is considered as a secondary winding of a transformer, where a separate primary winding is tightly coupled to secondary with negligible leakage (coupling coefficient K_{ps} close to unity). The primary to secondary turns ratio is typically 1:1 (or higher).
- ❑ All primary windings are connected in a series loop with an additional compensation inductor L_c , and grounded properly.
- ❑ Each magnetic core can have only one primary winding and one secondary winding for one phase (Figure a)
- ❑ Each magnetic core can have one primary winding and multiple secondary windings for multiple phases (Figure b). In this case, the inter-secondary coupling coefficient K_{ss} is much lower than K_{ps} . For example, a 6-phase VR can use 6 magnetic cores based on single-secondary structure, or use 3 magnetic cores based on dual-secondary structure.

Source: "Fast multi-phase trans-inductor voltage regulator", https://www.tdcommons.org/cgi/viewcontent.cgi?article=3261&context=dpubs_series

TLVR ELECTRICAL MODEL

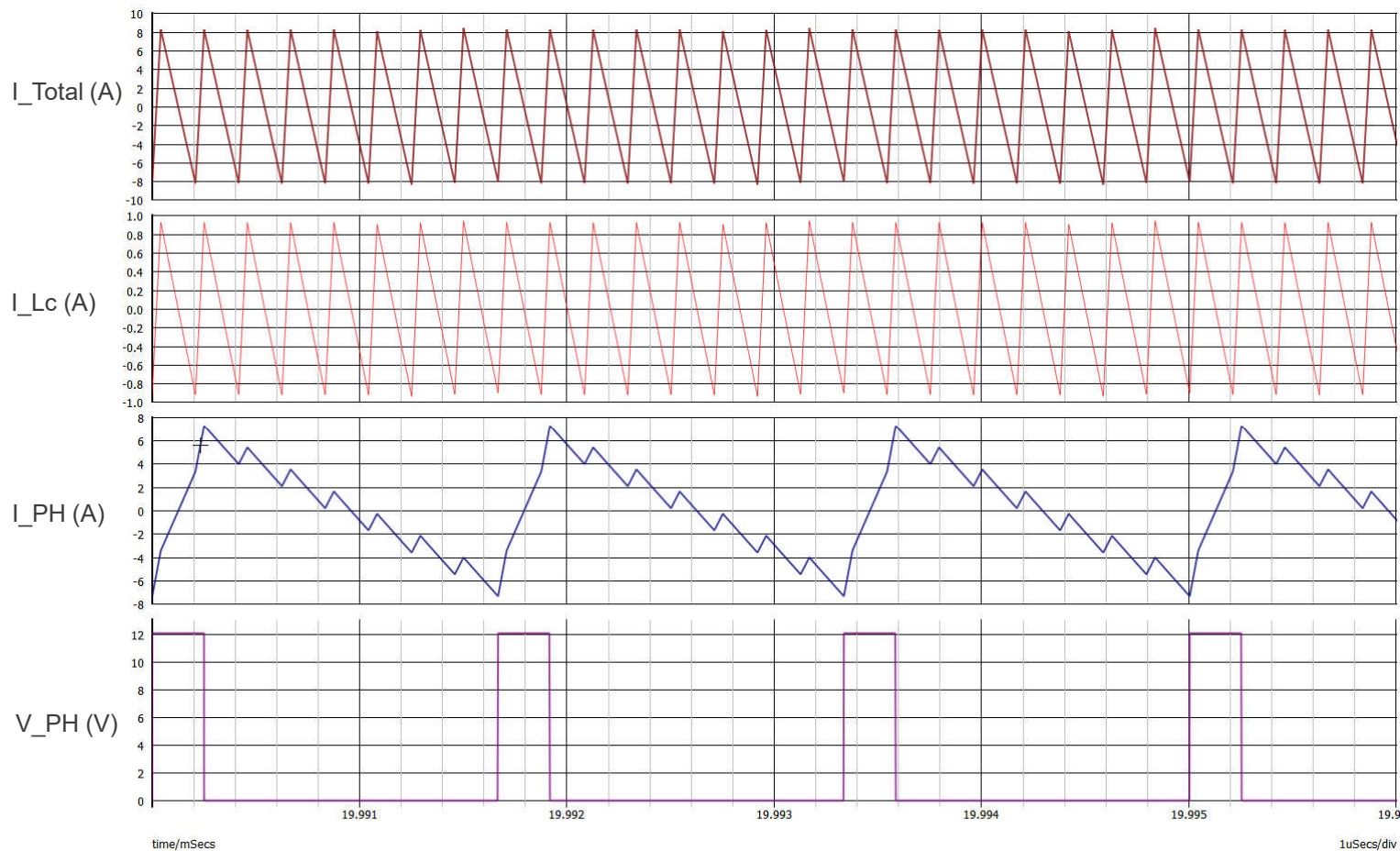


Source: "Fast multi-phase trans-inductor voltage regulator", https://www.tdcommons.org/cgi/viewcontent.cgi?article=3261&context=dpubs_series

TLVR SWITCHING WAVEFORMS

Test Conditions

Vin=12V, Vout=1.8V (~15% Duty), 8PH
Fsw=600kHz, L=200nH, Lc=150nH, k=0.98
Load = 0A



Total Current (sum of all phases)
ripple current = $16.7A_{pkpk}$

Coupling Inductor (Lc) ripple
current = $1.87A_{pkpk}$

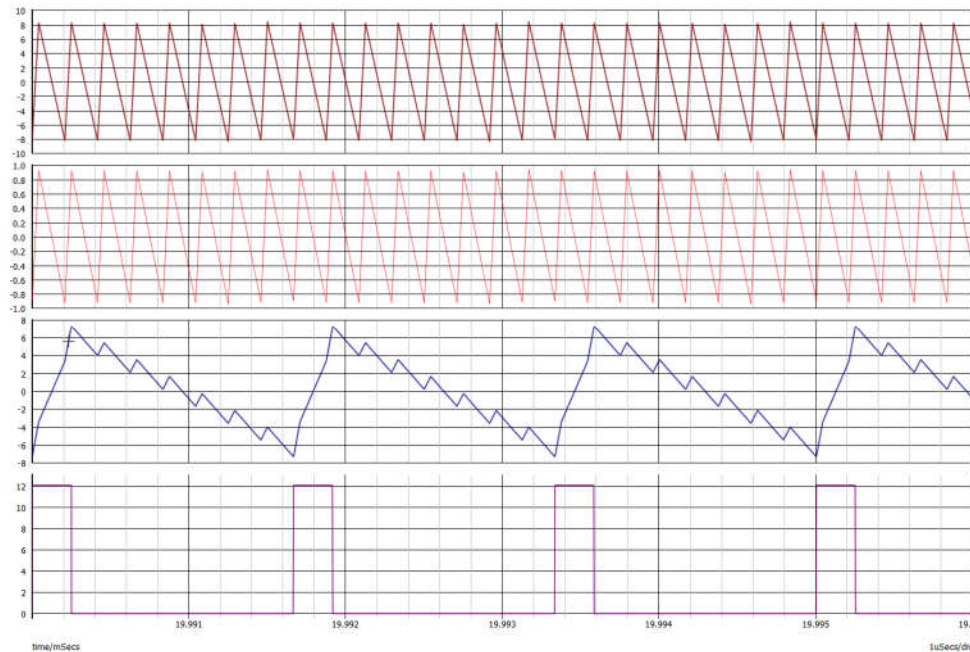
Per phase ripple current
= $14.53A_{pkpk}$

TLVR VS STANDARD L SWITCHING WAVEFORMS

Test Conditions

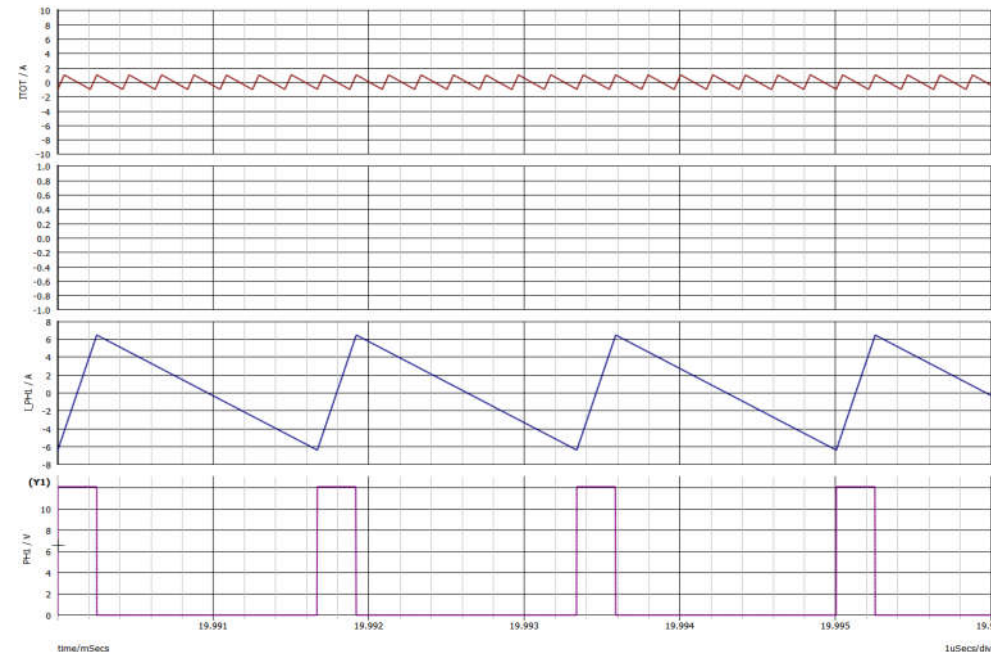
Vin=12V, Vout=1.8V (~15% Duty), 8PH
Fsw=600kHz, L=200nH, Lc=150nH, k=0.98
Load = 0A

TLVR L=200nH, Lc=150nH



— = Total current, 16.7A_{pkpk} @ Fsw*N
— = Lc current, 1.9A_{pkpk} @ Fsw*N
— = Phase current, 14.5A_{pkpk}

Standard L=200nH



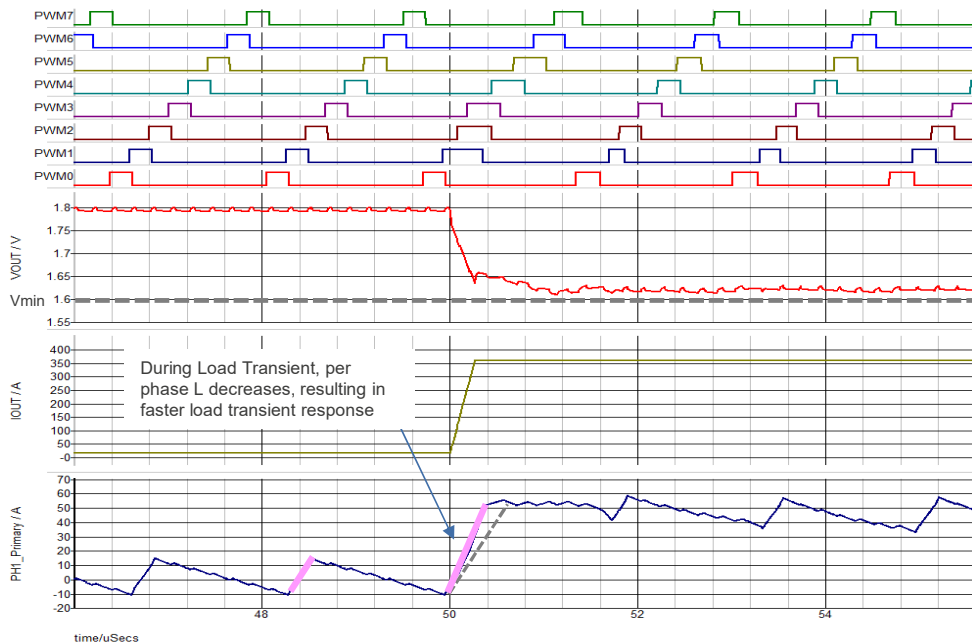
— = Total current, 2.0A_{pkpk} @ Fsw*N
— = Phase current, 12.7A_{pkpk}

TLVR VS STANDARD L SWITCHING WAVEFORMS

Test Conditions

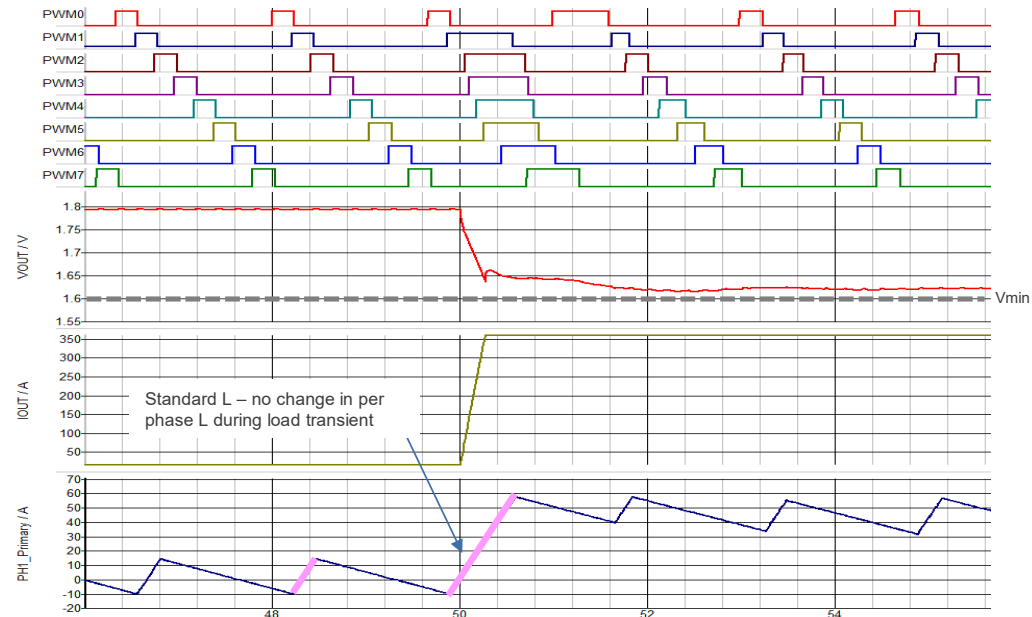
Vin=12V, Vout=1.8V (~15% Duty), 8PH
Fsw=600kHz, L=105nH, Lc=150nH, k=0.98
Load Step = 15A to 359A

TLVR L=105nH, Lc=150nH



— = Vout
— = Load Current
— = Phase current

Standard L=105nH



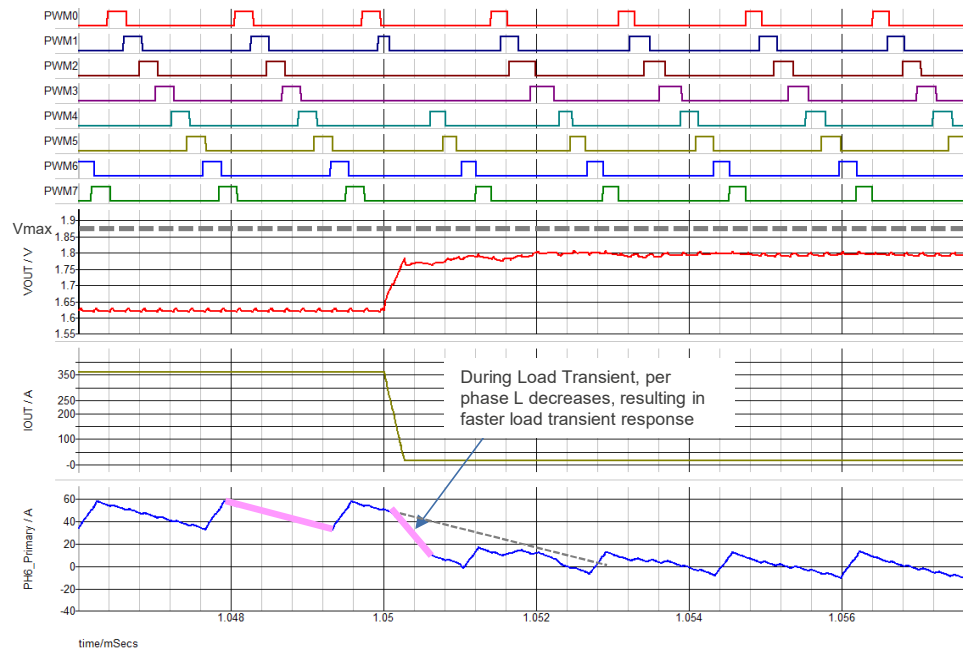
— = Vout
— = Load Current
— = Phase current

TLVR VS STANDARD L SWITCHING WAVEFORMS

Test Conditions

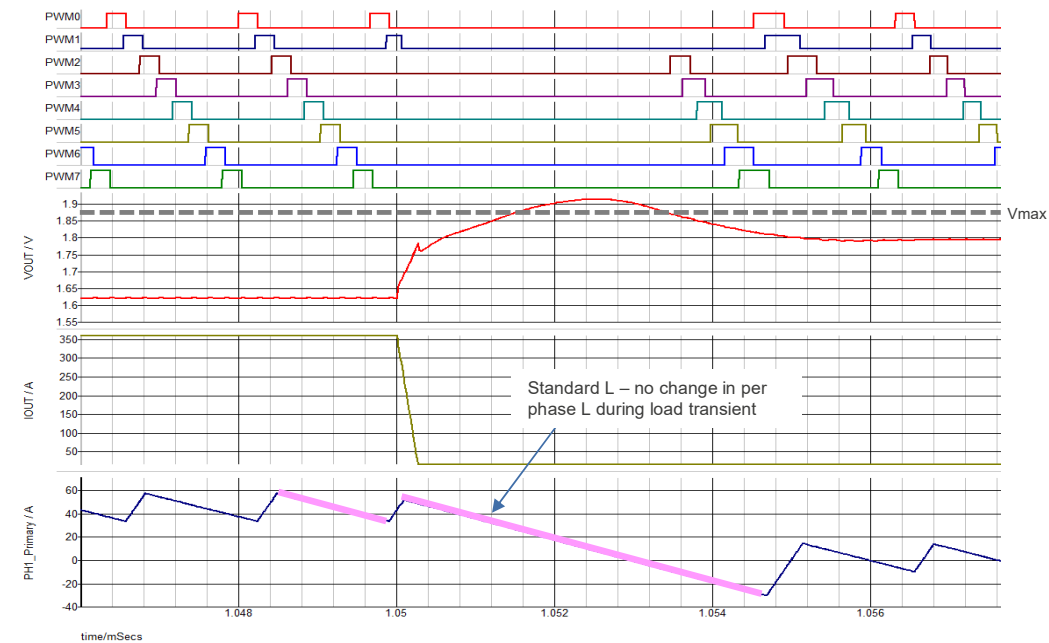
Vin=12V, Vout=1.8V (~15% Duty), 8PH
Fsw=600kHz, L=105nH, Lc=150nH, k=0.98
Load Release = 359A to 15A

TLVR L=105nH, Lc=150nH



— = Vout
— = Load Current
— = Phase current

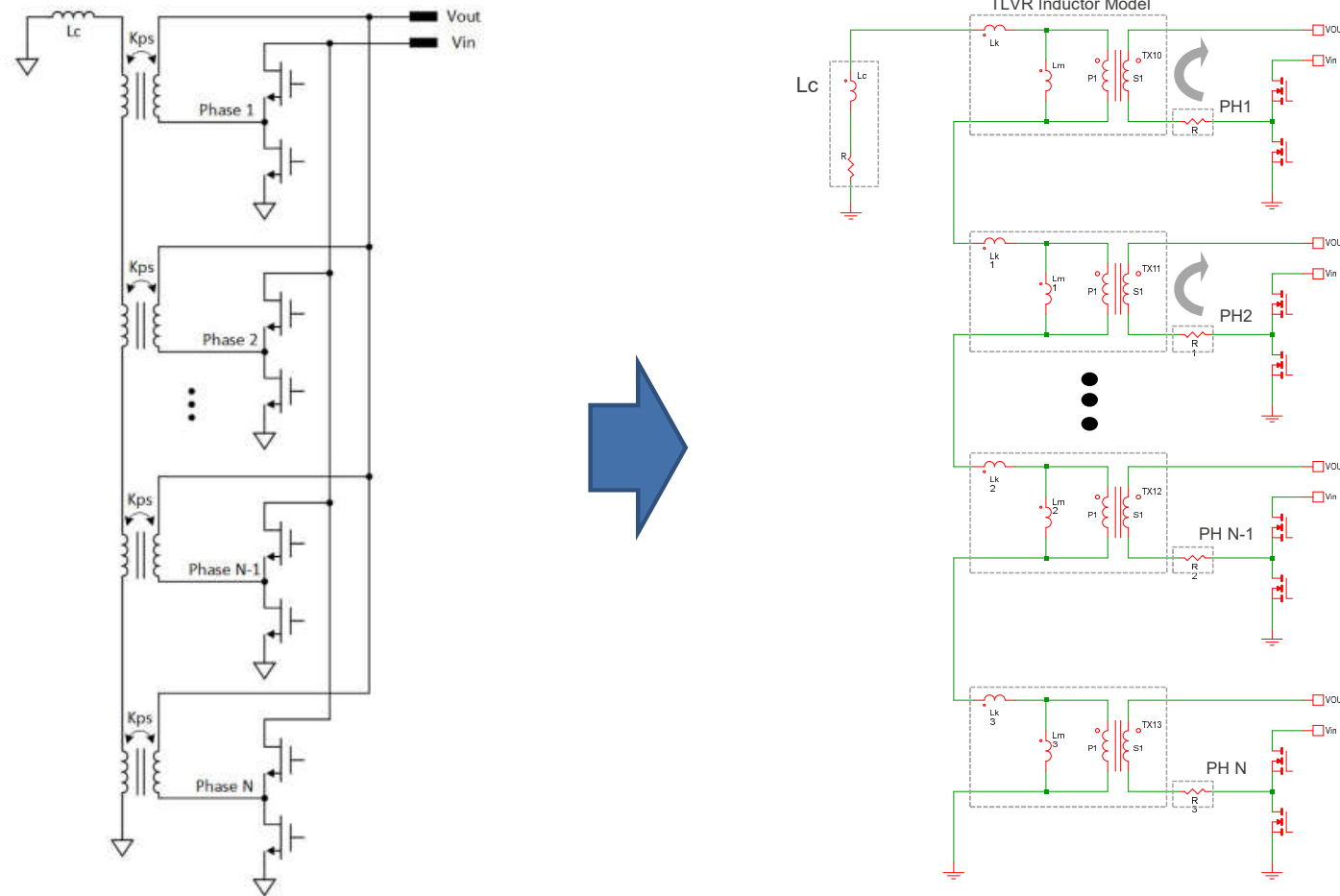
Standard L=105nH



— = Vout
— = Load Current
— = Phase current

TLVR DESIGN EQUATIONS

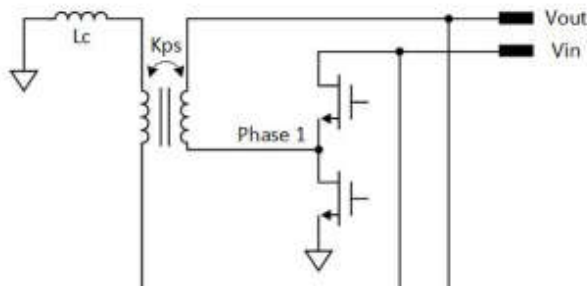
TLVR EQUIVALENT CIRCUIT MODELING



Note: In SIMPLIS, a non-ideal transformer is modeled by adjusting the coupling factor in the transformer model. L_k and L_m are not shown on the schematic.

TLVR EQUIVALENT CIRCUIT MODELING

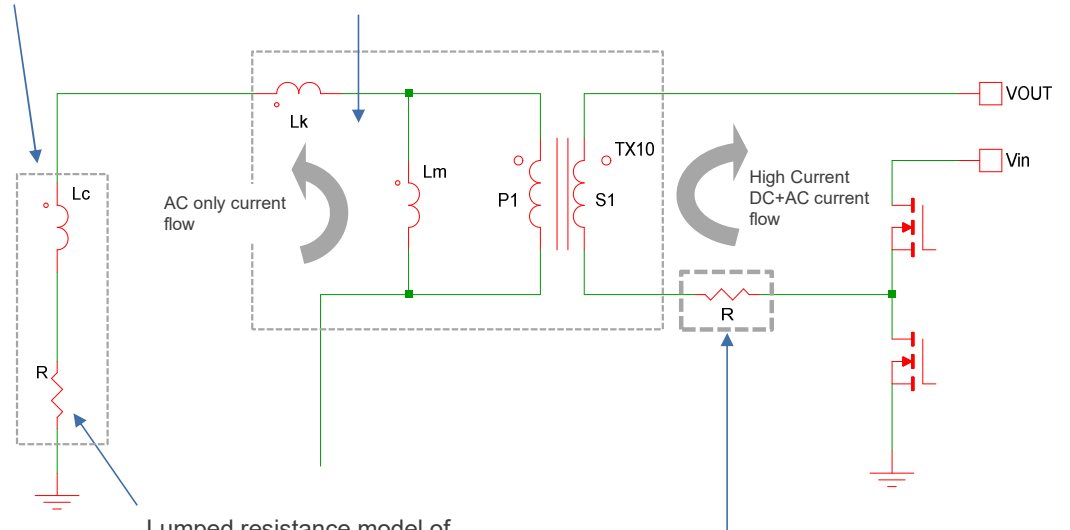
Circuit model from TLVR disclosure



Non ideal transformer model

- L_k = Leakage Inductance
- L_m = Magnetizing Inductance
- T_x = Ideal 1:1 Transformer

L_c = Coupling Inductor



Lumped resistance model of primary side and L_c . This side only sees AC current, so resistance can be higher.

Resistance of secondary side. This sees DC current, so must be low (typically $<0.2\text{m}\Omega$ for VR designs)

Note: In SIMPLIS, a non-ideal transformer is modeled by adjusting the coupling factor in the transformer model. L_k and L_m are not shown on the schematic.

TLVR EQUATIONS – LC RIPPLE CALCULATION

Lc Ripple Current Calculation:

$$I_{Cpkpk} = \frac{V_{in} \cdot \left((m + 1) - D \cdot N \right) \cdot \left(D - \frac{m}{N} \right)}{L_{eq} \cdot f_{sw}}$$

Where:

$$L_{eq} = (1 - k^2) \cdot LP \cdot N + Lc$$

$$D = \frac{V_{out}}{V_{in}}$$

$$m = \text{floor}\left(N \cdot \frac{V_{out}}{V_{in}}\right)$$

Note: The floor function is used to determine if there is PWM pulse overlap. If 1 or higher, PWM overlaps. If 0, no PWM overlap.

Example:

N=8PH, Vin=12V, Vout=1.8V, Lp=200nH, k=0.98,
Lc=150nH, Fsw=600kHz, Ts=1.667us

$$m = \text{floor}\left(8 \cdot \frac{1.8}{12}\right) = \text{floor}(1.2) = 1 \quad \text{Phases overlap}$$

$$L_{eq} = (1 - 0.98^2) \cdot 200nH \cdot 8 + 150nH = 213nH$$

$$D = \frac{1.8}{12} = 0.15$$

$$I_{Cpkpk} = \frac{12 \cdot \left((1 + 1) - 0.15 \cdot 8 \right) \cdot \left(0.15 - \frac{1}{8} \right)}{213nH \cdot 600kHz} = 1.88A_{pkpk}$$

Sim Result: $I_{Cpkpk} = 1.9A_{pkpk}$

TLVR EQUATIONS – PHASE RIPPLE CALCULATION

Output Phase Ripple

$$I_{phpkpk} = I_{cpkpk} + I_{Mpkpk}$$

Where:

$$I_{cpkpk} = \frac{V_{in} \cdot \left((m + 1) - D \cdot N \right) \cdot \left(D - \frac{m}{N} \right)}{L_{eq} \cdot f_{sw}}$$

$$I_{Mpkpk} = \frac{(V_{in} - V_{out}) \cdot V_{out}}{V_{in} \cdot L_p \cdot f_{sw}}$$

See previous slide for I_{cpkpk} calculation details

Example:

N=8PH, V_{in} =12V, V_{out} =1.8V, L_p =200nH, k =0.98,
 L_c =150nH, F_{sw} =600kHz, T_s =1.667us

$$I_{cpkpk} = \frac{12V \cdot \left((1 + 1) - 0.15 \cdot 8 \right) \cdot \left(0.15 - \frac{1}{8} \right)}{213nH \cdot 600kHz} = 1.88Apkpk$$

$$I_{Mpkpk} = \frac{(12V - 1.8V) \cdot 1.8V}{12V \cdot 200nH \cdot 600kHz} = 12.75Apkpk$$

$$I_{phpkpk} = 1.88Apkpk + 12.75Apkpk = \boxed{14.63Apkpk}$$

Sim Result: $I_{phasepkpk} = 14.5A_{pkpk}$

TLVR EQUATIONS – TOTAL CURRENT RIPPLE CALCULATION

I_{out} Total Ripple (sum of all phase current)

$$I_{Total} = \left(\frac{N \cdot V_{in}}{f_{sw}} \right) \cdot \left(D - \frac{m}{N} \right) \cdot \left(\frac{(1 + m)}{N} - D \right) \cdot \left(\frac{k \cdot N}{L_{eq}} + \frac{1}{L_p} \right)$$

Where:

$$L_{eq} = (1 - k^2) \cdot L_P \cdot N + L_c$$

$$D = \frac{V_{out}}{V_{in}}$$

$$m = \text{floor} \left(N \cdot \frac{V_{out}}{V_{in}} \right)$$

Example:

N=8PH, Vin=12V, Vout=1.8V, Lp=200nH, k=0.98,
Lc=150nH, Fsw=600kHz, Ts=1.667us

$$m = \text{floor} \left(8 \cdot \frac{1.8}{12} \right) = \text{floor}(1.2) = 1 \quad \text{Phases overlap}$$

$$L_{eq} = (1 - 0.98^2) \cdot 200nH \cdot 8 + 150nH = 213nH$$

$$D = \frac{1.8}{12} = 0.15$$

$$I_{Total} = \left(\frac{8 \cdot 12V}{600kHz} \right) \cdot (0.025) \cdot (0.1) \cdot \left(\frac{0.98 \cdot 8}{213nH} + \frac{1}{200nH} \right)$$

$$I_{Total} = \boxed{16.7A_{pkpk}}$$

Sim Result: I_{TOTpkpk} = 16.7A_{pkpk}

See previous slide for I_{c_{pkpk}} calculation details

TLVR EQUATIONS – EQL. L DURING A LOAD TRANSIENT

Equivalent L during a Load Transient*

$$L_T = L_{eq} \cdot \left(\frac{L_p}{L_c + N \cdot L_p} \right)$$

Where:

$$L_{eq} = (1 - k^2) \cdot L_p \cdot N + L_c$$

Example:

N=8PH, Vin=12V, Vout=1.8V, Lp=200nH, k=0.98,
Lc=150nH, Fsw=600kHz, Ts=1.667us

$$L_{eq} = (1 - 0.98^2) \cdot 200nH \cdot 8 + 150nH = 213nH$$

$$L_T = 213nH \cdot \left(\frac{200nH}{150nH + 8 \cdot 200nH} \right)$$

$$L_T = \boxed{24.3nH}$$

*Equivalent per phase inductance when all phases are either ON or OFF in response to a load transient

TLVR DESIGN PROCEDURE

TLVR DESIGN PROCEDURE

- 1) For a given family elect the largest TLVR inductance that still meets ISAT requirements.
- 2) Select Lc such that ripple and overshoot requirements are met.
 - 1) Transient requirements are limited by overshoot. Select Lc so that overshoot requirements are met.
 - 2) Next, verify resulting Vout ripple is within spec.
- 3) After selecting Lc, verify TLVR ISAT still meets requirements.
- 4) If Lc cannot be set low enough to meet transient requirements or TLVR ISAT is too low, adjust TLVR inductance down to next available value and recalculate Lc.

Example:

Vin = 12V, Vout = 1.83V
Load Line = 0.5mOhm
Istep = 15A to 359A @ 1050A/us
Imax = 450A
Target Fsw = 600kHz
8PH design

Vout Ripple < 10mV_{pkpk}
Max overshoot = 1.9V

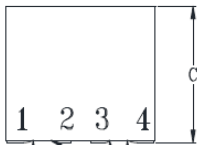
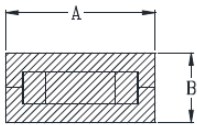
What TLVR inductance and Lc should be selected?

STEP 1: TLVR INDUCTANCE SELECTION

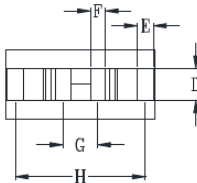
Delta DHCME1211D series...

Electrical Characteristics @ 25°C, 100kHz, 1V

Delta P/N	L (nH) ± 15%	Li (nH) MIN	DCR_P (mΩ) ± 10%	DCR_S (mΩ) ± 10%	Isat ¹ (A)		Ir_P ² (A)	Ir_S ² (A)
					25°C	100°C		
DHCME1211D-11168	105	74			100	85	77	35
DHCME1211D-12168	120	84			88	75		
DHCME1211D-15168	150	105	0.125	0.60	68	58		
DHCME1211D-16168	160	112			63	54		
DHCME1211D-17168	170	119			60	51		
DHCME1211D-20168	200	140			50	42		



DCR_S Measure Points
DCR_P Measure Points



$$IL_{sat} > \frac{IMAX}{N} + \frac{1}{2} I_{ph} = \frac{450}{8} + \frac{1}{2} * 20 = 66.3A$$

Estimated per phase ripple current

Initial TLVR L selection is 120nH based on estimated Isat requirements.

STEP 2: LC SELECTION, TRANSIENT

Lc selection using Vout overshoot:

$$L_{TMax} \leq 0.9 \cdot \frac{2 * N * C * Vo}{I_{step}^2} \cdot \left[\Delta V_{max} - \left(I_{step} \cdot ESR + ESL \cdot \frac{I_{step}}{T_{rise}} \right) \right]$$

↑
Overshoot from inductor
energy dumped into Cout
↑
Overshoot from
parasitic ESR
↑
Overshoot from
parasitic ESL

↙ 2nd Droop contribution
to load step

$$L_C \leq \frac{L_P \cdot N \cdot (L_{TMax} - (1 - k^2) \cdot LP)}{L_P - L_{TMax}}$$

Where

$$\Delta V_{max} = V_{max} - (V_{out} - (I_{max} * LoadLine))$$

C = Total output cap with derating

ESR and ESL can be approximated by simulating PDN and measuring results R and L at $\frac{0.35}{RT_{ISTEP}}$

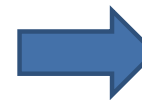
STEP 2: LC SELECTION, TRANSIENT

All ceramic caps should be de-rated to account for DC Bias, Temperature and Tolerance effects. A good rule of thumb is ~50% derating. Any bulk capacitors should be de-rated by ~20%.

	Cout (uF)	Qty	Size	Actual (uF)	Derating Factor	Derated (uF)
C_Bulk	None	--	--	--	--	--
C_cer_inductor	22uF	8	0805	176uF	50%	88uF
C_midplane	100uF	24	0805	2400uF	50%	1200uF
C_socket_bottom	100uF 22uF	26 6	0805 0603	2600uF 132uF	50%	1300uF 66F
C_socket_top	22uF	80	0402	1760uF	50%	880uF

Total:

7,068uF



3,543uF

STEP 2: LC SELECTION, TRANSIENT

Lc selection using Vout overshoot:

$$\Delta V_{max} = 1.8V - (1.8V - (359A * 0.5m\Omega)) = 180mV$$

$$L_{TMax} \leq 0.9 \cdot \frac{2 * 8 * 3534\mu F * 1.8V}{(344A)^2} * \left[180mV - \left(344A * 0.33m\Omega + 17pH * \frac{344}{265ns} \right) \right]$$

$$L_{TMax} \leq 34.25nH$$

$$Lc \leq \frac{200nH \cdot 8 \cdot (34.25 - (1 - 0.95^2) \cdot 200nH)}{200nH - 34.25nH}$$

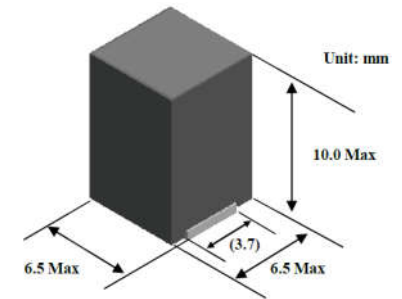
$$Lc \leq \boxed{252nH}$$

LC INDUCTANCE SELECTION

LC ISAT must be high enough to handle dynamic current during load step

Electrical Specification

TDK Identification	Inductance		Test Freq. (kHz)	DC Resistance		Rated DC Current (A)					Status
	at 0A (uH)	Tol. (%)		Spec. Typ (m-Ohm)	Tol. (%)	Isat_20% 25°C	Typ. 100°C	Isat_30% 25°C	Typ. 100°C	Itemp_40C TYP.	
VLBU6565100T-R07L	0.07	+/-15%	100	0.17	+/-10%	110	94	115	97	70	Sam.
VLBU6565100T-R10L	0.10	+/-15%	100	0.17	+/-10%	80	70	85	73	70	M.P.
VLBU6565100T-R12L	0.12	+/-15%	100	0.17	+/-10%	69	60	72	62	70	Sam.
VLBU6565100T-R15L	0.15	+/-15%	100	0.17	+/-10%	52	46	56	49	70	M.P.
VLBU6565100T-R18L	0.18	+/-15%	100	0.17	+/-10%	39	37	42	38	70	Sam.
VLBU6565100T-R20L	0.20	+/-15%	100	0.17	+/-10%	37	34	41	36	70	M.P.
VLBU6565100T-R22L	0.22	+/-15%	100	0.17	+/-10%	33	30	36	31	70	M.P.
VLBU6565100T-R33L	0.33	+/-15%	100	0.17	+/-10%	20	18	22	20	70	Sam.



Initial TLVR Lc selection is 150nH based on L_{TMax} calculations and required ISAT.

STEP 2: LC SELECTION, VOUT RIPPLE

Vout Ripple Estimation:

$$V_{out_ripple} = \underset{\substack{\uparrow \\ \text{Output Cap ESR} \\ \text{Contribution}}}{I_{tot}pk_{pk} * ESR} + \underset{\substack{\uparrow \\ \text{Output Cap ESL} \\ \text{Contribution}}}{\frac{I_{tot}pk_{pk}}{T_{on}} * ESL} \quad \text{where } T_{on} = T_s * \left[D - \frac{m}{N} \right]$$

$$ESR \text{ at } N * f_{sw} \approx 0.37m\Omega$$

$$ESL \text{ at } N * f_{sw} \approx 19pH$$

$$m = \text{floor}\left(N \cdot \frac{V_{out}}{V_{in}}\right)$$

To estimate ESR and ESL, simulate output cap and PCB impedance profile, then measure ESR and ESL at N*Fsw

Selecting Lc = 150nH...

$$V_{out_ripple} = 15.8Apkpk * 0.37m\Omega + \frac{15.8Apkpk}{45.8ns} * 19pH = \boxed{12.7mVpkpk}$$

STEP 3: ISAT VERIFICATION

TLVR ISAT Check:

$$I_{phpkpk} = I_{Cpkpk} + I_{Mpkpk} = 1.64A + 21.25A = 22.9A_{pkpk}$$

$$ISAT > \frac{IMAX}{N} + \frac{1}{2}(I_{PHpkpk}) = \frac{450A}{8} + \frac{1}{2}(22.9A)$$

$$ISAT > 67.7A$$

Selected ISAT is higher than max required, selected inductor will work in this application!

Electrical Characteristics @ 25°C, 100kHz, 1V

Delta P/N	L (nH) ± 15%	Li (nH) MIN	DCR_P (mΩ) ± 10%	DCR_S (mΩ) ± 10%	Isat ¹ (A)		Ir_P ² (A)	Ir_S ² (A)
					25°C	100°C		
DHCME1211D-11168	105	74	0.125	0.60	100	85	77	35
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DHCME1211D-15168	150	105			68	58		
DHCME1211D-16168	160	112			63	54		
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DHCME1211D-20168	200	140			50	42		

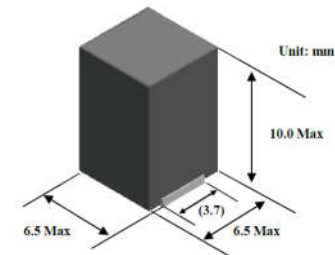
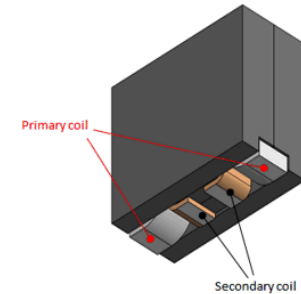
STEP 4: DESIGN SUMMARY

Component Selection:

- TLVR = 120nH Delta DHCME1211D
- Lc = 150nH TDK VLBU6565120T
- Fsw = 600kHz

Design Estimates:

- Lc Ripple = 1.64A
- PH Ripple = 22.9A
- I_{Total} Ripple = 15.8A
- Estimated Vout Ripple = 12.8mVpkpk
- Estimated Vout Overshoot = <1.800V
- Estimated 2nd Droop = 135mV



TLVR DUAL SOURCING AND LAYOUT

TLVR INDUCTOR SUPPLIERS

Renesas is engaged with multiple vendors on TLVR inductors:

1) Delta – Samples received and tested 

2) TDK – Samples received and tested 

3) ITG – Samples received, preliminary testing complete

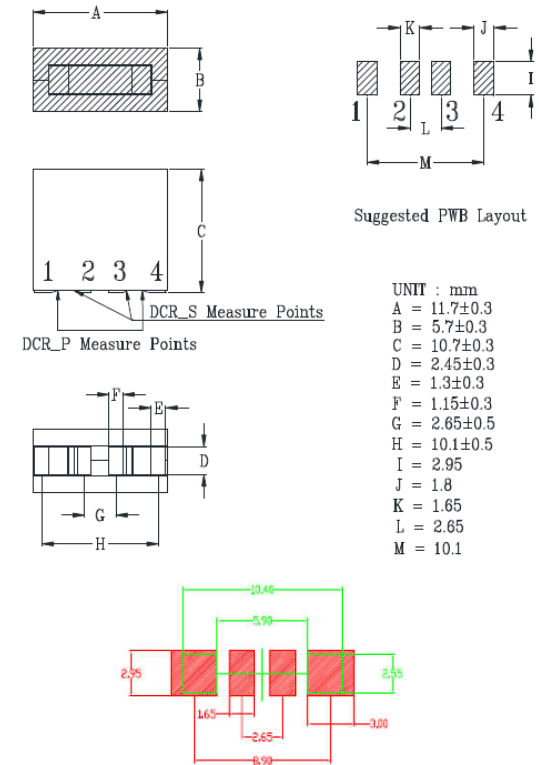


4) Eaton – Samples Pending 

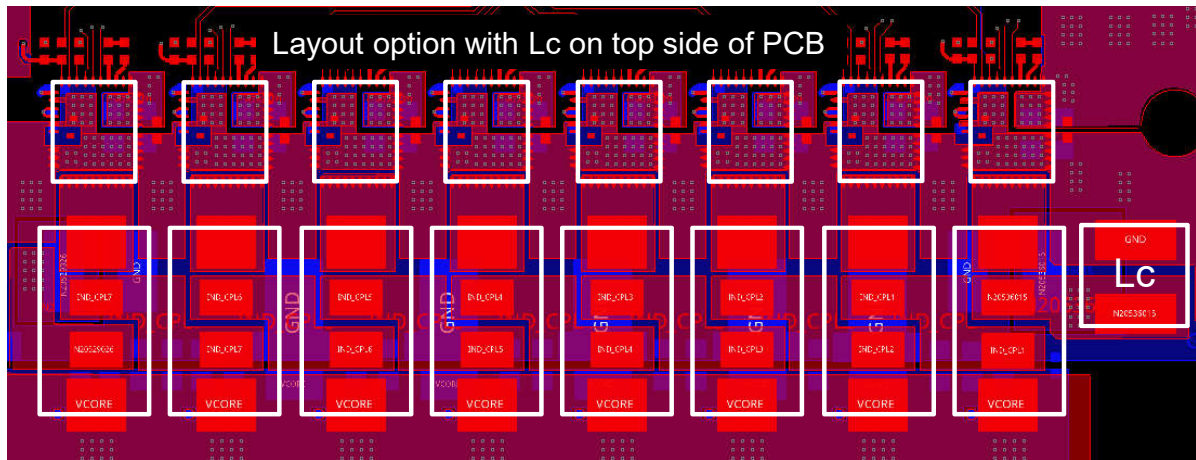
TLVR INDUCTOR FOOTPRINT 1

Initial TLVR footprint is in a 6x12mm (WxL) footprint. For risk mitigation, standard inductor dual footprint solutions also exist.

- Delta – DHCME1211D Series
Standard L Dual Footprint Option: HCBS9610F
- TDK – VLBC12060120 Series
Standard L Dual Footprint Option: VLBU12060120

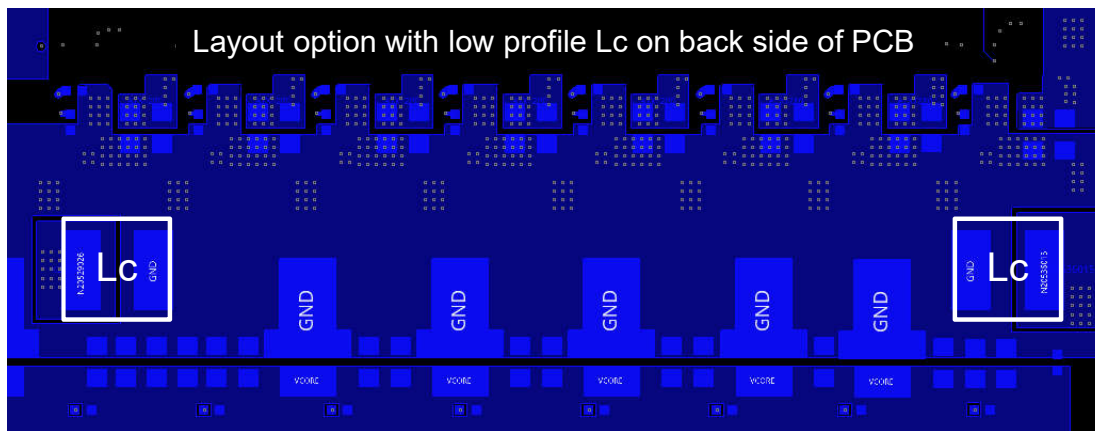


TLVR INDUCTOR FOOTPRINT 1 EXAMPLE LAYOUT



Layout Option w/ LC on top side of PCB

- LxW, excluding Lc: 23x60mm
- LxW including Lc: 23x68mm
- Lc Example: VLBU6565100T
 - 6.5x6.5x10mm
 - Hard saturation Ferrite type
 - L=150nH, ISAT=52A, DCR=0.17mΩ



Layout Option w/ LC on bottom side of PCB

- Minimal increase in solution width, LxW including Lc: 23x61mm
- Dual Lc option
- Lc Example: SMA2509A
 - 7.3x6.8x2.4mm
 - Soft saturation molded type L=100nH, ISAT=50A, DCR=0.8mΩ

BIG IDEAS FOR EVERY SPACE

Renesas.com